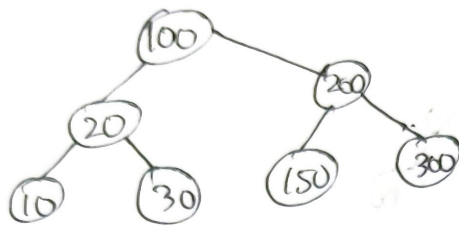


cos-AT3 Tree Traversal Challenges

1)



$$\text{In order (L, N, R)} = \underline{10, 20, 30} \quad \underline{100} \quad \underline{150, 200, 300}$$

L N R

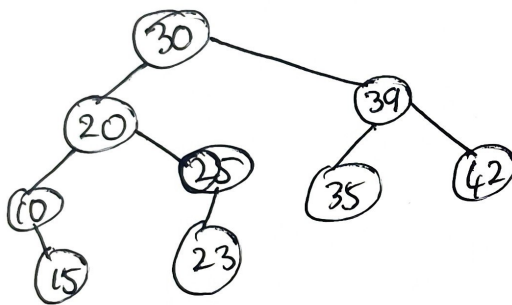
$$\text{Preorder (Node, L, R)} = \underline{100} \quad \underline{20, 10, 30} \quad \underline{200, 150, 300}$$

N L R

$$\text{Postorder (L, R, N)} = \underline{10, 30, 20} \quad \underline{150, 300, 200} \quad \underline{100}$$

L R N

2)



$$\text{Inorder (L, N, R)} = \underline{10, 15, 20, 23, 25} \quad \underline{30} \quad \underline{35, 39, 42}$$

L N R

$$\text{Preorder (N, L, R)} = \underline{30} \quad \underline{20, 10, 15, 25, 23} \quad \underline{39, 35, 42}$$

N L R

$$\text{Postorder (L, R, N)} = \underline{15, 10, 23, 25, 20} \quad \underline{35, 42, 39} \quad \underline{30}$$

L R N

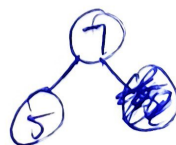
3) Binary Search tree - 7, 5, 1, 8, 3, 6, 0, 9, 4, 2

Insert 7



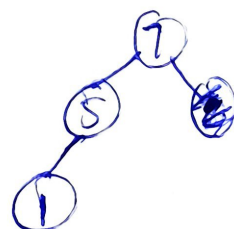
→

Insert 5

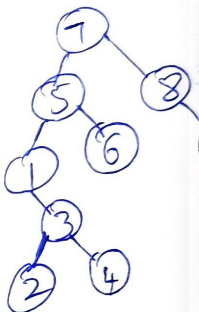


→

Insert 1



Insert 8



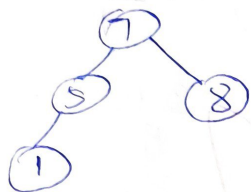
Inorder

Preorder

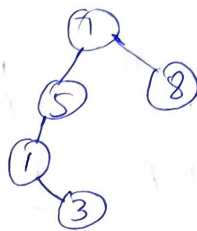
Postorder

5)

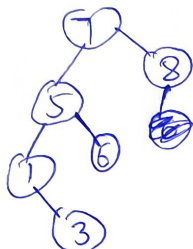
Insert 8



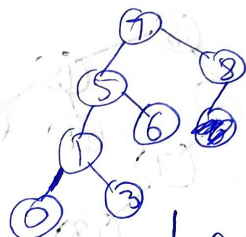
→ Insert 3



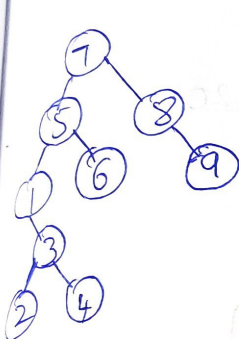
Insert 6



→ Insert 0

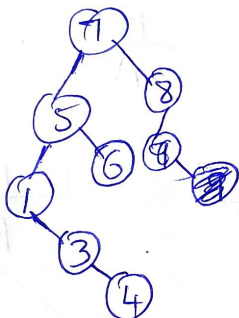


↓ Insert 9

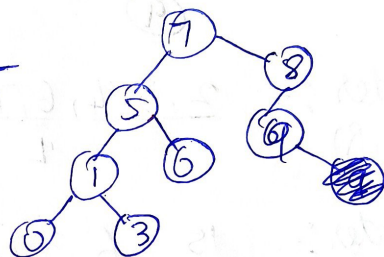


← 2

Insert 4



←



Inorder(L, N, R):-

0, 1, 2, 3, 4, 5, 6

7
N

8, 9
R

Preorder(N, L, R):-

7
N

5, 1, 0, 3, 2, 4, 6
L

8, 9
R

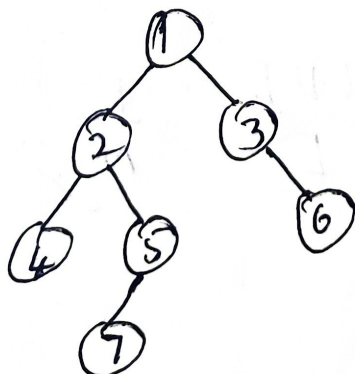
Postorder(L, R, N):-

0, 2, 4, 3, 1, 6, 5
L

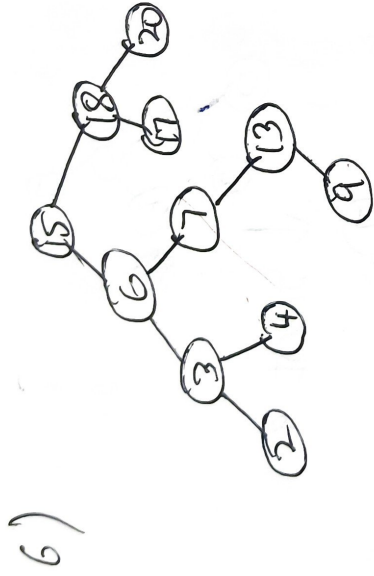
9, 8
R

7
N

5)



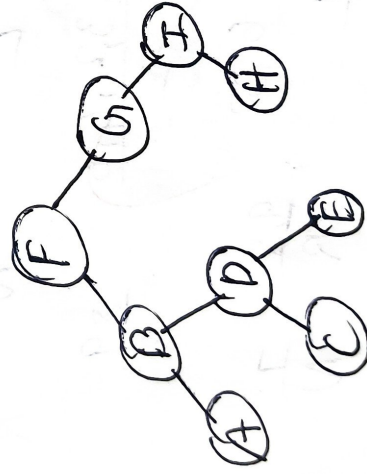
$\text{Inorder}(C, L, N, R) = \frac{4, 2, 7, 5}{L} \quad \frac{1}{N} \quad \frac{3, 6}{R}$
 $\text{Preorder}(N, L, R) = \frac{1}{N} \quad \frac{2, 4, 5, 7}{L} \quad \frac{3, 6}{R}$
 $\text{Postorder}(L, R, N) = \frac{4, 7, 5, 2}{L} \quad \frac{6, 3}{R} \quad \frac{1}{N}$



$\text{Inorder} :- \frac{2, 3, 4, 6, 7, 9, 13}{L} \quad \frac{15}{N} \quad \frac{17, 18, 20}{R}$

$\text{Preorder} :- \frac{15}{N} \quad \frac{6, 3, 2, 4, 7, 13, 9}{L} \quad \frac{18, 17, 20}{R}$

$\text{Postorder} :- \frac{2, 4, 3, 9, 13, 7, 6}{L} \quad \frac{17, 20, 18}{R} \quad \frac{15}{N}$

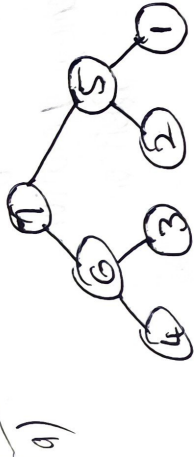


$\text{Inorder}(L, N, R) :- \frac{A, B, C, D, E}{L} \quad \frac{F}{N} \quad \frac{G, H, I}{R}$

8) Inorder: D, B, E, F, A, G, H, C
 Preorder: A, B, D, E, F, C, G, H
 (N, L, R)

Post order (L, R, N):- D, F, E, B, H, G, C, A

$\Rightarrow$

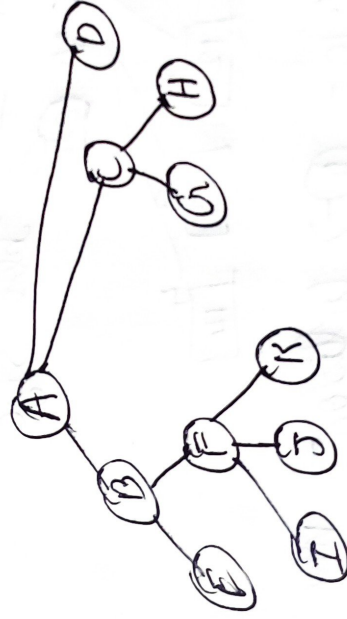


Inorder: (L, N, R) : $\frac{4, 6, 3}{L} \cdot \frac{7}{N} \cdot \frac{2, 5, 1}{R}$

Preorder: (N, L, R) $\frac{7}{N} \cdot \frac{6, 4, 3}{L} \cdot \frac{2, 5, 1}{R}$

Postorder: (L, R, N) : $\frac{4, 3, 6}{L} \cdot \frac{2, 5, 1}{R} \cdot \frac{7}{N}$

10)



Preorder: (N, L, R) $\frac{A}{N} \cdot \frac{B, E, F, I, J, K}{L} \cdot \frac{C, D, H, P}{R}$

~~Inorder: (L, R, N) $\frac{E, I, J, K, F, B, G, H, C, D, A}{L} \cdot \frac{A}{N} \cdot \frac{C, D, H, P}{R}$~~

Post order: (L, R, N) : $\frac{E, I, J, K, F, B, G, H, C, D}{L} \cdot \frac{A}{N} \cdot \frac{C, D, H, P}{R}$

Level-order: $A, B, C, D, E, F, G, H, I, J, K$
 (Left to Right).